KOLEJ MATRIKULASI KELANTAN

SECTION A [25 marks]

This section consists of **3** questions. Answers **all** the questions given.

1. Evaluate the following limits:

(a) $\lim\limits_{x \to 2} \dfrac{x^3 - 8}{x^2 - 2x}$.

[3 marks]

(b) $\lim\limits_{x \to \infty} \sqrt{\dfrac{5x+7}{6x-5}}$.

[3 marks]

(c) $\lim\limits_{x \to 3^-} \dfrac{|4x-12|}{x-3}$.

[3 marks]

2. (a) Given $g(x) = (2-x)^3(x-1)$. Find $g'(x)$ in factored form.

[3 marks]

(b) Given $y = e^{-2x} \sin 3x$. Find $\dfrac{dy}{dx}$ and $\dfrac{d^2y}{dx^2}$.

Hence, show that $\dfrac{d^2y}{dx^2} + 4\dfrac{dy}{dx} + 13y = 0$

[6 marks]

3. The surface area of a balloon in the shape of a sphere is decreasing at the rate of $2\ \text{cm}^2/\text{min}$. Find the rate at which the volume is decreasing when the radius of the balloon is $5\,\text{cm}$.

[7 marks]

SECTION B [75 marks]

This section consists of **7** questions. Answers **all** the questions given.

1. Given a complex number $z = 2 + i$.

   (a) Express $\bar{z} - \frac{1}{\bar{z}}$ in the form $a + bi$, where a and b are real numbers.

   [3 marks]

   (b) Obtain $\left| \bar{z} - \frac{1}{\bar{z}} \right|$.

   [2 marks]

2. (a) Solve the equation $3\log_9 x = (\log_3 x)^2$.

   [7 marks]

   (b) Solve $\frac{5}{x+3} - 1 \leq 0$.

   [4 marks]

3. Given the function $g(x) = \frac{1}{2x-5}$.

   (a) Show that $g(x)$ is a one-to-one function. Hence, find $g^{-1}(x)$.

   [5 marks]

   (b) Show that $g \circ g^{-1}(x) = x$.

   [2 marks]

4. Consider functions of $f(x) = (x-2)^2 + 1,\ x > 2$ and $g(x) = \ln(x+1), x > 0$.

   (a) Find $f^{-1}(x)$ and $g^{-1}(x)$. State the domain and range for each of the inverse function.

   [9 marks]

   (b) Obtain $(g \circ f)(x)$. Hence, evaluate $(g \circ f)(2)$.

   [4 marks]

5. State the conditions for continuity of $f(x)$ at $x = a$.

[2 marks]

(a) By using conditions for continuity of $f(x)$ at $x = a$, find the values of m and n such that $f(x) = \begin{cases} n - 2\cos x, & x < 0 \\ 2 + mx^2, & 0 \le x < 2 \\ m - x, & x \ge 2 \end{cases}$ is continuous on the interval $(-\infty, \infty)$.

[7 marks]

(b) If $m = -2$ and $n = 4$, determine whether $f(x)$ is differentiable at $x = 2$ or not.

[5 marks]

6. (a) A curve is defined by the parametric equations $x = 3t - \frac{1}{t}$ and $y = t + \frac{3}{t}$ where $t \neq 0$. Show that $\frac{dy}{dx} = \frac{t^2 - 3}{3t^2 + 1}$. Hence, find $\frac{d^2y}{dx^2}$.

[8 marks]

(b) A curve with equation $x^2 - 3y^2 = ae^{y-2x} + by - 6$, where a and b are contants, passes through the point (1,2). Given $\frac{dy}{dx} = 1$ at $(1, 2)$, determine the values a and b.

[8 marks]

7. The function f is defined by $f(x) = \frac{\ln(x-1)}{x-1}$ for $x > 1$.

(a) By considering the first and second derivatives of $f(x)$, show that there is only one maximum point on the graph $y = f(x)$

[7 marks]

(b) Use the result in part **7**(a) to state the exact coordinates of the maximum point.

[2 marks]

END OF QUESTION PAPER

Final Answer

SECTION A

1. (a) 6

   (b) $\sqrt{\frac{5}{6}}$

   (c) -4

2. (a) $(2-x)^2(5-4x)$

   (b) $\frac{dy}{dx}=e^{-2x}(-2\sin 3x+3\cos 3x)$

   $\frac{d^2y}{dx^2}=-5e^{-2x}\sin 3x-12e^{-2x}\cos 3x$

3. $-5\text{ cm}^3/\text{min}$

SECTION B

1. (a) $\frac{8}{5}-\frac{6}{5}i$

   (b) 2

2. (a) $x=1$, $x=3^{\frac{3}{2}}$

   (b) $(-\infty,-3)\cup[2,\infty)$

3. (a) $\frac{1+5x}{2x}$

   (b) Show

4. (a) $f^{-1}(x)=\sqrt{x-1}+2$, $g^{-1}(x)=e^x-1$

   $D_{f^{-1}}=\{x>1\}, R_{f^{-1}}=\{y>2\}$

   $D_{g^{-1}}=\{x>0\}, R_{g^{-1}}=\{y>0\}$

   (b) $\ln[(x-2)^2+2]$, $\ln 2$

5 $(i)\, f(a)\ is\ defined$

$(ii)\lim_{x\to a} f(x) is\ exists, \lim_{x\to a^-} f(x)=\lim_{x\to a^+} f(x)$

$(iii)\, f(a)=\lim_{x\to a} f(x)$

   (a) $n=4\quad m=\frac{-4}{3}$

(b) $f(x)$ *is not differentiable at* $x = 2$

6. (a) Show, $\dfrac{d^2y}{dx^2} = \dfrac{20t^3}{\left(3t^2+1\right)^3}$

(b) $a = 5,\ b = -5$

7. (a) Show

(b) $\left(e+1, \dfrac{1}{e}\right)$